Supermarket Sales Analysis – Results and Insights

1. Introduction

Supermarkets generate large amounts of transactional data every day. Analyzing this data helps businesses understand customer behavior, product performance, and sales patterns. Data analysis can reveal insights that support better business decisions, improve customer satisfaction, and increase profitability.

This project focuses on analyzing a supermarket sales dataset using Python, Pandas, Matplotlib, and Jupyter Notebook. The objective is to explore the dataset, perform data cleaning, analyze patterns in sales and customer ratings, and visualize key insights.

2. Dataset Overview

The dataset contains **2000 sales records with 14 columns** describing supermarket transactions.

Dataset Shape:
Rows: 2000
Columns: 14

Dataset Columns

Column	Description
Invoice_ID	Unique transaction ID
Branch	Store branch where the transaction occurred
City	City where the branch is located
Customer_Type	Type of customer (Member / Normal)
Gender	Gender of the customer
Product_Line	Category of products sold
Unit_Price	Price per product unit
Quantity	Number of products purchased

Tax	Tax applied to the purchase
Total	Total bill amount
Date	Date of transaction
Time	Time of transaction
Payment	Payment method used
Rating	Customer satisfaction rating

The dataset provides useful information for understanding sales distribution, product popularity, and customer satisfaction.

3. Data Cleaning and Preprocessing

Before performing analysis, the dataset was cleaned to ensure accuracy and consistency.

Steps Performed

1. **Checked dataset structure**

   ```
   df.shape
   df.columns
   df.head()
   ```

2. **Checked for missing values**

   ```
   df.isnull().sum()
   ```

   No major missing values were found in the dataset.

3. **Converted Date column to datetime format**

   ```
   df["Date"] = pd.to_datetime(df["Date"])
   ```

   This allows time-based analysis and trend visualization.

4. **Checked data types**

```
df.dtypes
```

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns in the dataset using statistical summaries and visualizations.

Key aspects analyzed include:

- Customer satisfaction ratings
- Product category performance
- Sales distribution
- Customer purchase behavior
- Payment methods

5. Average Rating by Product Line

To understand customer satisfaction across product categories, the average rating was calculated for each product line.

```
top_rated_products =  df.groupby("Product_Line")["Rating"].mean().sort_
values(ascending=False)

print(top_rated_products)
```

Result

Product Line	Average Rating
Food & Beverages	7.05
Health & Beauty	6.99
Home & Lifestyle	6.97
Fashion Accessories	6.94
Sports & Travel	6.93
Electronic Accessories	6.86

6. Visualization

A **bar chart** was created to visualize the average rating by product line.

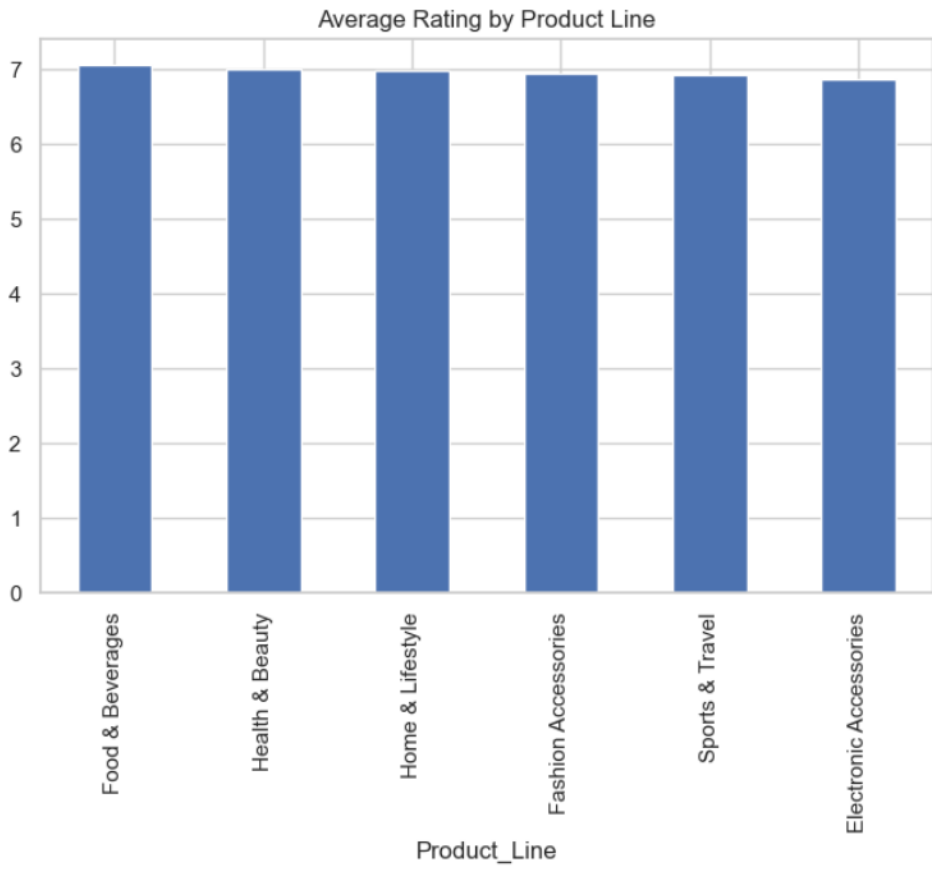

Figure 1: Average Rating by Product Line

7. Interpretation of Results

The analysis shows that Food & Beverages received the highest average customer rating (7.05). This indicates that customers are generally more satisfied with products in this category.

On the other hand, **Electronic Accessories received the lowest average rating (6.86)** among all product categories, suggesting slightly lower customer satisfaction compared to other product lines.

Although differences in ratings exist, the ratings are relatively close, ranging between **6.86 and 7.05**, which shows that overall customer satisfaction is consistent across product categories.

8. Additional Possible Visualizations

To gain deeper insights, additional visualizations can be included such as:

Sales by City

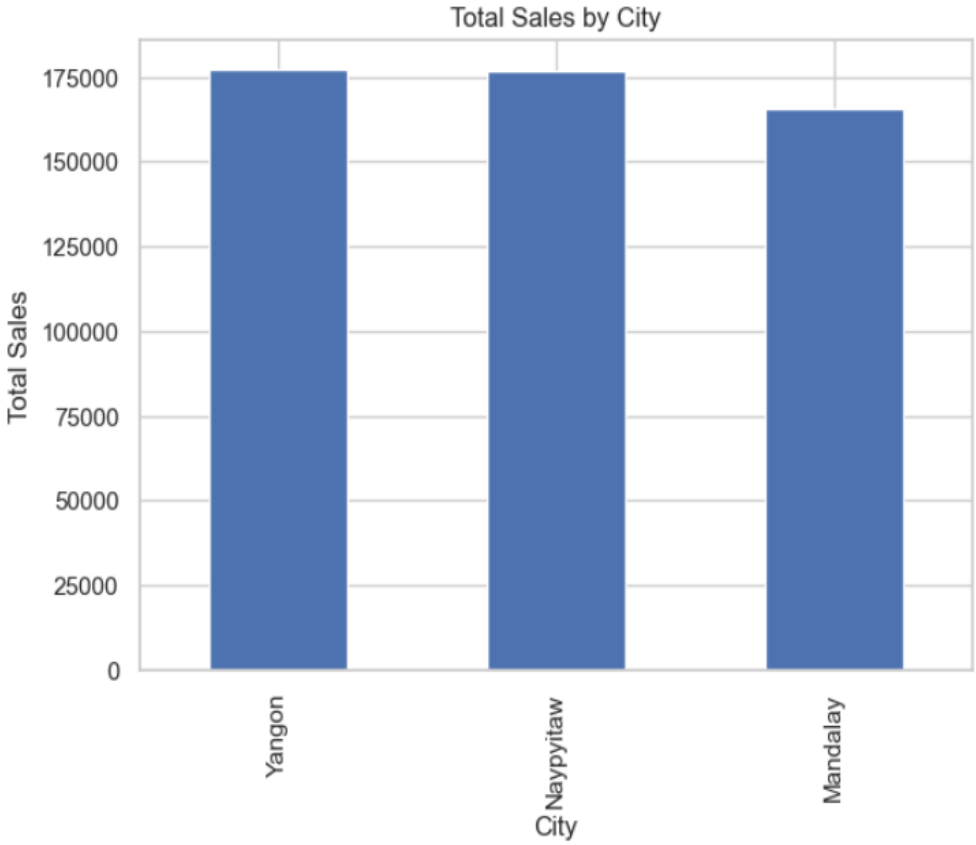

Figure 2: Sales Distribution by City

Sales by Product Line

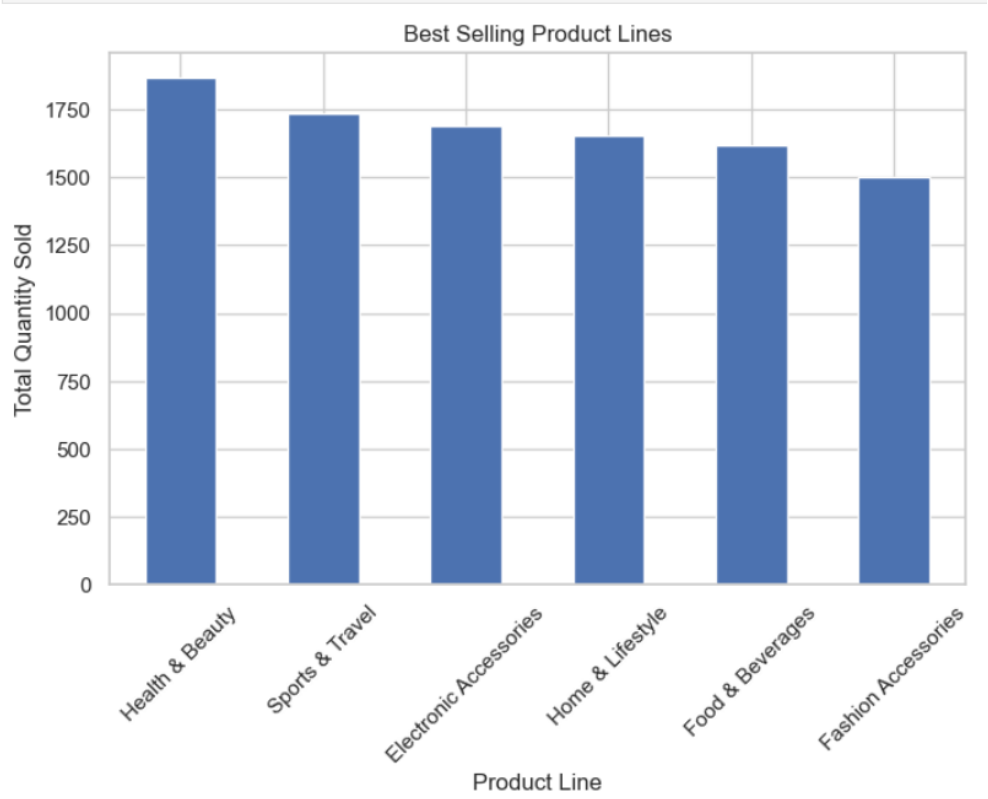

Figure 3: Sales by Product Category

Payment Method Distribution

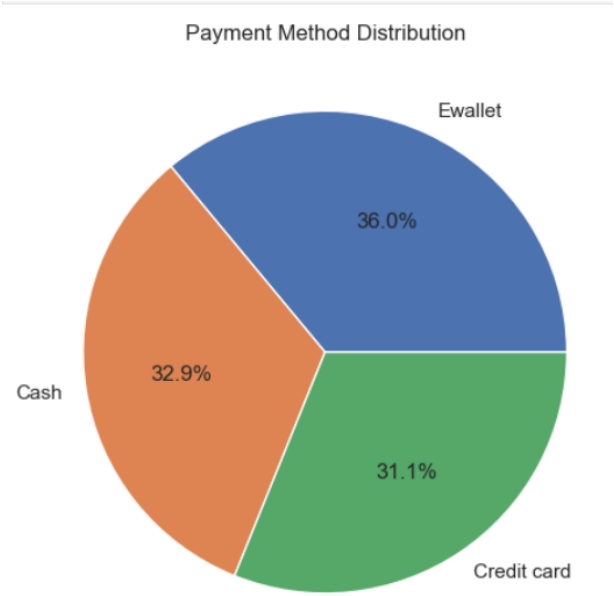

Figure 4: Payment Method Usage

Quantity Purchased Distribution

Figure 5: Quantity Distribution

9. Key Insights

From the analysis, several insights were identified:

- Food & Beverages has the highest customer satisfaction rating.
- Electronic Accessories has slightly lower ratings compared to other categories.
- Customer ratings across product lines are generally consistent.
- Sales data can help supermarkets understand customer preferences and improve product offerings.

These insights can help businesses make data-driven decisions to improve customer experience and product quality.

10. Conclusion

This project analyzed supermarket sales data using Python-based data analysis tools. The dataset was cleaned, explored, and visualized to uncover patterns related to customer ratings and product performance.

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Data analysis like this helps businesses improve product quality, enhance customer satisfaction, and make better strategic decisions.